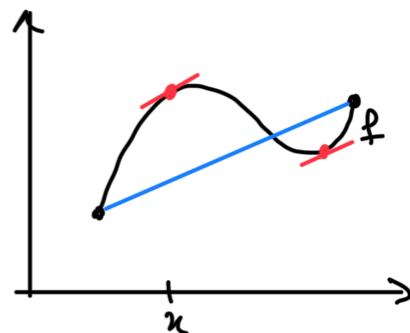


Thm (Mean value theorem) If $f: [a, b] \rightarrow \mathbb{R}$ is continuous and f is differentiable on (a, b) , then $\exists \xi \in (a, b)$ s.t.

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$



Thm (Taylor's theorem with remainder) Let $I \subseteq \mathbb{R}$ be an open interval, $f: I \rightarrow \mathbb{R}$, and $x_0 \in I$. If the n th derivative $f^{(n)}: I \rightarrow \mathbb{R}$ of f exists, then $\forall x \in I \setminus \{x_0\} \exists y$ between x_0 and x s.t.

$$f(x) = \underbrace{\sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k}_{(n-1)\text{st Taylor polynomial approximation of } f} + \underbrace{\frac{f^{(n)}(y)}{n!} (x-x_0)^n}_{\text{remainder}}.$$

Pf: Fix $x \in I \setminus \{x_0\}$ and $n \geq 1$. Pick $M \in \mathbb{R}$ s.t.

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + \frac{M}{n!} (x-x_0)^n.$$

Want: $M = f^{(n)}(y)$ for some $y \in \begin{cases} (x_0, x) & \text{if } x > x_0, \\ (x, x_0) & \text{if } x < x_0. \end{cases}$

Set

$$g(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (t-x_0)^k + \frac{M}{n!} (t-x_0)^n - f(t).$$

Then:

$$g(x) = 0 \quad (\text{by our choice of } M)$$

$$g(x_0) = \frac{f(x_0)}{1!} + 0 + \dots + 0 + 0 - f(x_0) = 0$$

$$\stackrel{MVT}{\Rightarrow} \exists x_1 \text{ between } x_0, x \text{ s.t. } g'(x_1) = 0$$

$$\begin{aligned} g'(x_0) &= \frac{d}{dt} \left[f(x_0) + f'(x_0)(t-x_0) + \dots + \frac{M}{n!}(t-x_0)^n - f(t) \right]_{t=x_0} \\ &= 0 + f'(x_0) + 0 + \dots + 0 - f'(x_0) = 0 \end{aligned}$$

$$\stackrel{MVT}{\Rightarrow} \exists x_2 \text{ between } x_0, x_1 \text{ s.t. } g''(x_2) = 0$$

$$\begin{aligned} g''(x_0) &= \frac{d^2}{dt^2} \left[f(x_0) + f'(x_0)(t-x_0) + \frac{f''(x_0)}{2}(t-x_0)^2 + \dots - f(t) \right]_{t=x_0} \\ &= 0 + 0 + 2 \cdot 1 \cdot \frac{f''(x_0)}{2} + 0 + \dots + 0 - f''(x_0) = 0 \end{aligned}$$

$$\stackrel{MVT}{\Rightarrow} \dots$$

Repeating this n times, we get:

$$\exists x_n \text{ between } x_0 \text{ and } x_{n-1} \text{ s.t. } g^{(n)}(x_n) = 0.$$

So:

$$\begin{aligned} 0 &= g^{(n)}(x_n) \\ &= \frac{d^n}{dt^n} \left[f(x_0) + \dots + \frac{f^{(n-1)}(x_0)}{(n-1)!}(t-x_0)^{n-1} + \frac{M}{n!}(t-x_0)^n - f(t) \right]_{t=x_n} \\ &= 0 + \dots + 0 + \frac{M}{n!} \cdot n \cdot (n-1) \cdot \dots \cdot 1 - f^{(n)}(x_n) \\ &= M - f^{(n)}(x_n) \end{aligned}$$

$$\Rightarrow M = f^{(n)}(x_n), \quad \text{with } x_n \text{ between } x_0 \text{ and } x,$$

as desired. □

Cor 1 Moreover, if also $f^{(n)}: I \rightarrow \mathbb{R}$ is bounded, then $\exists C_n > 0$ s.t.

$$\underbrace{\left| f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k \right|}_{\text{the error in the approximation}} \leq \frac{C_n}{n!} \underbrace{|x-x_0|^n}_{\text{is higher order}} \quad \forall x \in I.$$

Ex $f(x) = e^x$

Suppose we want to approximate $f(1) = e$.

We know $e < 3$, and so $1 \in (-R, R)$ for $R = \ln 3$.

Then $f^{(n)}: (-R, R) \rightarrow \mathbb{R}$, $f^{(n)}(x) = e^x$ is bounded by $e^R = 3 \quad \forall n$. So, by Corollary 1,

$$\left| f(1) - \sum_{k=0}^{n-1} \frac{f^{(k)}(0)}{k!} 1^k \right| \leq \frac{3}{n!} \cdot 1^n$$

$$\Rightarrow \left| e - \sum_{k=0}^{n-1} \frac{1}{k!} \right| \leq \frac{3}{n!}$$

This gives us a way to approximate e and guarantee that the approximation is as accurate as we want, without having to know the value of e beforehand.

Suppose we want an approximation of e within 10^{-3} . Pick n s.t.

$$\frac{3}{n!} \leq 10^{-3} \quad \Leftrightarrow \quad n=7 \text{ works}$$

Then

$$\left| e - \underbrace{\sum_{k=0}^6 \frac{1}{k!}}_{= \frac{1957}{720}} \right| \leq \frac{3}{n!} \leq 10^{-3}$$
$$= \frac{1957}{720} = \underline{2.71805}$$

(Compare: $e = 2.71828\dots$)

Goal: When can we differentiate/integrate power series?

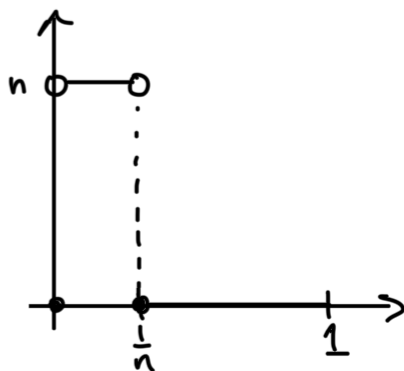
Q: When does

$$\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b \lim_{n \rightarrow \infty} f_n ?$$

A: Not always!

Ex $f_n: [0, 1] \rightarrow \mathbb{R}$,

$$f_n(x) = \begin{cases} n & \text{if } x \in (0, \frac{1}{n}), \\ 0 & \text{otherwise.} \end{cases}$$



Then:

$$\textcircled{1} \int_0^1 f_n = n \cdot \frac{1}{n} = 1 \quad \forall n$$

$\textcircled{2}$ f_n converges pointwise to $f(x) = 0$.

So

$$\int_0^1 f_n = 1 \not\rightarrow 0 = \int_0^1 f.$$

• So "converges pointwise" is not strong enough.